Needs-based recommendation systems

Machine Learning Pipeline for Financial Needs Prediction and Product Recommendation

Authors: Peng Rao, Maxime d'Aviau de Ternay, Qiao Zhaochen, Shourya Vashistha, Arnau Nomen

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1. Exploratory Data Analysis
2. Data Preparation & Feature Engineering
3. ML Workflow & Hyperparameter Optimization & Explainable AI (XAI)
4. Deep Learning Approach
5. Recommendations

Project Flow

1. Exploratory Data Analysis
Data quality, distributions, outliers

↓

2. Classification & Model Selection
Baselines, tuning, explainability, DCN comparison

↓

3. Recommendation Engine
SVD/Autoencoder ranking and product matching

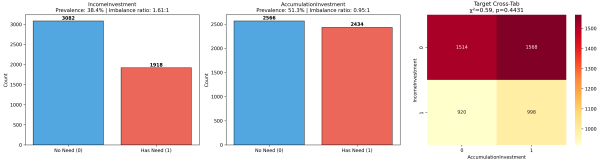
Target Variable Analysis

Key Findings

- `AccumulationInvestment`: **balanced** (51.3%, $\approx 1{:}1$)
- `IncomeInvestment`: **moderate imbalance** (38.4%, $\approx 1.6{:}1$)
- χ^2 test: $p = 0.44 \gg 0.05 \Rightarrow$ targets are independent.

Modeling Implication

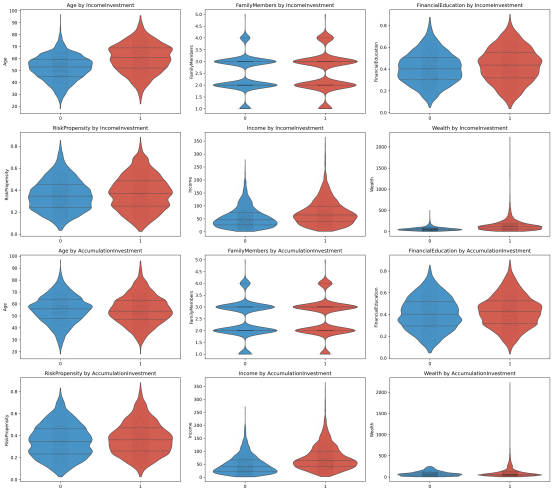
Naive all-zero baseline gives $\sim 62\%$ accuracy, so accuracy alone is misleading. We optimize $\mathbf{F}_{0.5}$ and model the two targets as outputs.



Target prevalence and cross-tab evidence

Feature Distributions by Target

- `Age`, `Income`, `Wealth`, `RiskPropensity`: distributions are **significantly higher for class 1**, indicating that these variables strongly discriminate between the two groups
- `FinancialEducation`: **moderate upward shift** for class 1, suggesting a secondary but positive effect
- `FamilyMembers`: **no visible difference**



Outlier Detection — Isolation Forest

Method choice

- **Isolation Forest** (300 estimators, `random_state=42`)
- Features standardised (StandardScaler) before fitting
- `Gender` (binary): inclusion does not distort the geometry
- Threshold: **5th percentile** of anomaly scores
  $\Rightarrow$ flags $\approx 5\%$ of observations as outliers

Decision: outliers retained

- Extreme `Income` / `Wealth` values are **valid** business profiles, not data errors
- Preferred treatment: **robust scaling** +

2D projections (Isolation Forest)

Key observations from scatter plots

Income vs Wealth: outliers concentrated at very high values — long right tail confirms extreme inequality.

Age vs Income / Wealth: older clients more likely to be flagged — plausible, not errors.

t-SNE projection: outliers (red) form a sparse fringe around the main cluster (blue). No clear isolated sub-group.

Conclusion: high-value financial profiles are structurally different, not corrupted data.

Feature Transformation and Engineering

1. Data Transformation (`DataTransformer`)

- **Numeric Conversion:** Coercion of mixed/string formats to continuous variables.
- **Log Transformation:** Applied $\log(1+x)$ to highly skewed financial variables (`Wealth`, `Income`) to normalize distributions.
- **Scaling:** `StandardScaler` applied post-engineering for stable convergence.

2. Household Metrics

- `WealthIncomeRatio`
- `IncomePerFamilyMember`

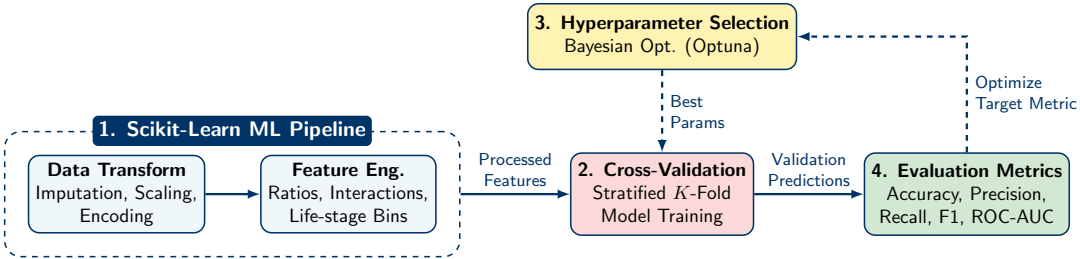
3. Interaction Features

- `RiskEducationInteraction` ($Risk \times Edu$)
- `RiskWealthInteraction` ($Risk \times \log(Wealth)$)
- `AgeRiskInteraction` ($Age \times Risk$)

4. Non-linearity & Life-Stages

- `AgeSquared`: Models the "Life-Cycle" curve (wealth peaking in mid-life).
- **Age Bins**: One-hot flags for `Under35`, `35_54`, `55_69`, and `70plus`.

ML Pipeline Overview



Baseline Models Evaluated

Models Evaluated

We comprehensively tested several families of classifiers across both target variables (*Income* and *Accumulation*):

- **Distance/Linear**: SVM, Naive Bayes, KNN.
- **Tree-based Ensembles**: Random Forest, XGBoost, LightGBM.
- **Meta-Ensembles**:
  - Voting Classifier (Soft & Hard)
  - Stacking Classifier (LGBM, XGB, KNN, RF with a Logistic Regression meta-learner)

Top Performers & Selection

Selected for Tuning:

- **Random Forest**
- **LightGBM**

Selection Rationale:

- Strong baseline predictive power with robust handling of non-linear interactions.
- **Target Metric:** Optimised for **F0.5 Score** to prioritise precision over recall (minimising the risk of unsuitable, false-positive product recommendations).

Optuna Hyperparameter Tuning

Objective Metric

- Optimized for **F0.5 Score** (Stratified 5-Fold CV)
- *Rationale:* Prioritizes precision over recall ($\beta = 0.5$), significantly penalizing false positives (bad recommendations).

TPE Sampler Strategy

- Uses Tree-structured Parzen Estimator (TPE).
- Explores the search space efficiently by modelling the probability of trial performance.

Hyperparameter Search Space

Random Forest (20 Trials)

- `n_estimators`: 10–200
- `max_depth`: 3–10
- `min_samples_split/leaf`: Constraints for regularization.

LightGBM (50 Trials)

- `learning_rate`: 0.001–0.2 (Log scale)
- `num_leaves`: 15–255
- `reg_alpha` & `reg_lambda`: L1/L2 Regularization (Log scale)

Optimization Results

- The tuned models demonstrated a significant reduction in false positives while maintaining competitive accuracy.

	Income Target			Accumulation Target		
Model	**Precision**	**Recall**	**F1 Score**	**Precision**	**Recall**	**F1 Score**
Baseline RF	0.7953	0.5260	0.6332	0.8301	0.7622	0.7947
Tuned RF	**0.8361**	0.5312	0.6497	**0.8581**	0.7544	0.8029
Baseline LGBM	0.8402	0.5339	**0.6529**	0.8503	0.7641	0.8049
Tuned LGBM	**0.8909**	0.5104	0.6490	**0.8568**	**0.7700**	**0.8111**

Final Selection

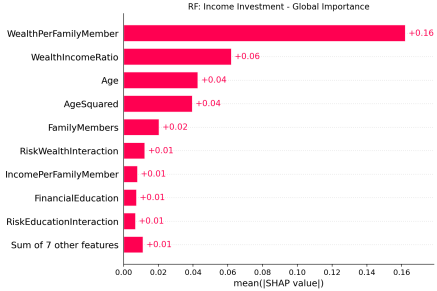
For the final recommendation pipeline, we use **Tuned RF** for **Income Investment** and **Tuned LightGBM** for **Accumulation Investment**, balancing precision and task-specific predictive strength.

XAI Methods: SHAP, LIME & Permutation

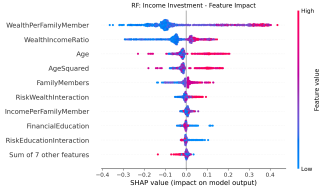
- **SHAP (SHapley Additive exPlanations):** Used to understand global feature importance and directionality across the entire dataset. It highlights which engineered features drive predictions the most.
- **LIME (Local Interpretable Model-agnostic Explanations):** Used to explain individual, client-level predictions to build trust and provide clear reasoning for a recommended product.

Model Insights: Income (RF) vs Accumulation (LGBM)

Income Investment

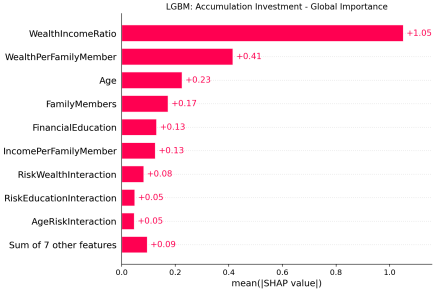


SHAP Feature Importance

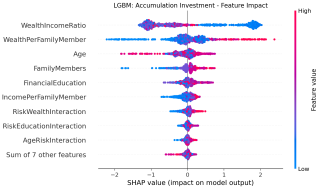


SHAP Beeswarm Plot

Accumulation Investment

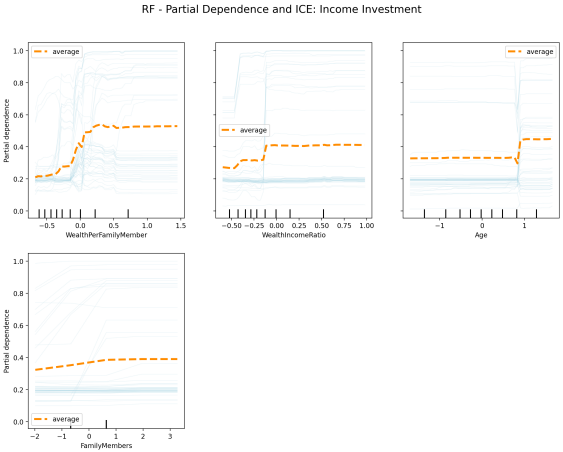


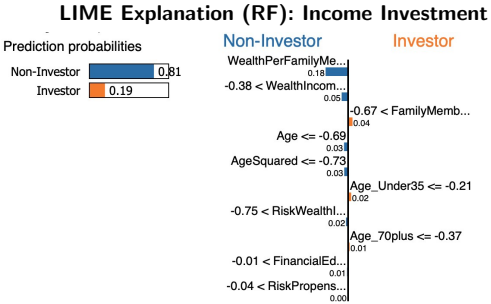
SHAP Feature Importance



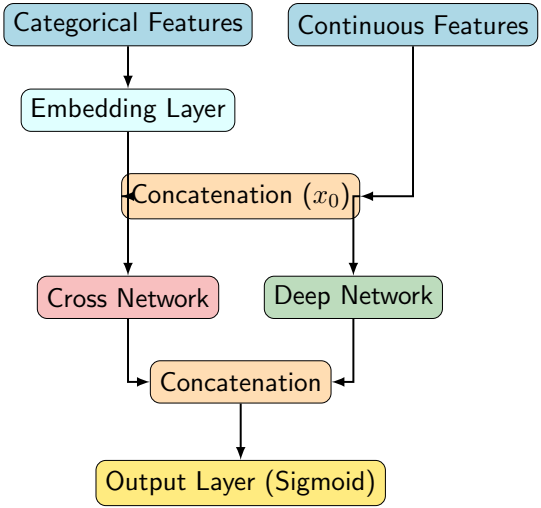
SHAP Beeswarm Plot

LIME, PDP and ICE for RF income investment





DCN Overview and Selection Decision [1]



DCN Results

Task	Acc	Prec	Rec	F1	AUC
Income	0.776	0.827	0.547	0.659	0.765
Accum.	0.795	0.851	0.702	0.769	0.848

Why Not Selected

- Slight gain for **Income** vs tuned RF for F1, but not for precision.
- Lower performance for **Accumulation** vs tuned LightGBM.
- Lower interpretability than tree-based models with SHAP/LIME.

Methodology: User-Item Matrix

We construct a user-item interaction matrix R using predicted probabilities from the supervised classifiers, penalised by risk misalignment.

Matrix Entry R_{ij}

$$R_{ij} = P_{\text{type}(j)}(i) \cdot \big(1 - \alpha \cdot |\text{RiskPropensity}(i) - \text{Risk}(j)|\big)$$

- $P_{\text{type}(j)}(i)$: Predicted probability of client i needing product j's type (Accumulation or Income).
- $\alpha \in [0, 1]$: Decay factor penalising risk misalignment.
  - **Economic Rationale**: Reflects the firm's risk aversion and regulatory suitability constraints (e.g., MiFID II).
  - **Trade-off**: Balances maximising expected fee revenue/conversion against minimising compliance risks and long-term customer churn from unsuitable investments.

Methodology: Models & Evaluation

We use model-based CF [2] leveraging matrix R:

- **Truncated SVD**: Factorises $R \approx U_k \Sigma_k V_k^T$ into low-dimensional latent factors to capture hidden preferences.
- **Autoencoder**: Learns non-linear representations by compressing and reconstructing the interaction matrix ($z_u = f_\theta(r_u)$, $\hat{r}_u = g_\phi(z_u)$).

Evaluation Metrics

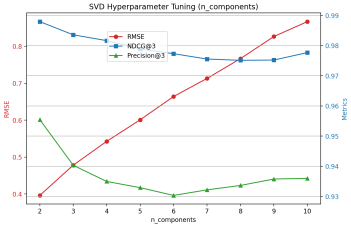
- **RMSE**: Measures the model's reconstruction error.
- **Precision@k & Recall@k**: Evaluate the relevance of the top-k recommendations.
- **NDCG@k**: Assesses ranking quality, prioritising relevant items placed higher.

Experimental Results

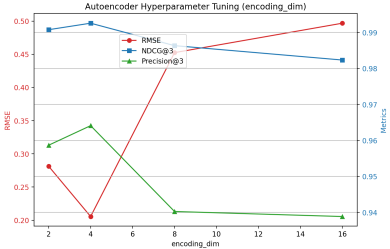
Table: *

Holdout validation results (production models: SVD n=6, AE dim=8)

Model	RMSE	Precision@3	Recall@3	NDCG@3
Truncated SVD	0.664	0.930	0.930	0.977
Autoencoder	**0.016**	**0.994**	**0.994**	**0.999**



SVD Tuning



Autoencoder Tuning

Financial Perspective

References

[1] Weiyu Guo et al. "Explainable enterprise credit rating using deep feature crossing". en. In: *Expert Systems with Applications* 220 (June 2023), p. 119704.

[2] Marwa Sharaf et al. "A survey on recommendation systems for financial services". In: *Multimedia Tools and Applications* 81.12 (May 2022), pp. 16761–16781.